

Industrial Internship Report on
"ServiceHub – Full Stack Service Marketplace Application"

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Executive Summary

My project was ServiceHub – Full Stack Service Marketplace Application.

This report presents the work completed during the Industrial Internship organized by upskill Campus and UniConverge Technologies Pvt. Ltd. (UCT). The internship provided an opportunity to apply theoretical knowledge to a real-world software development project.

The project undertaken during the internship was "ServiceHub – Full Stack Service Marketplace Application". The main objective of the project was to develop a centralized platform where customers can discover and book services while service providers can manage and promote their services through a web-based application.

The application was developed using the MERN Stack, including MongoDB, Express.js, React.js, and Node.js. Additional technologies such as JWT Authentication, REST APIs, GitHub, and MongoDB Atlas were used to build a secure and scalable solution.

This internship helped me gain practical experience in full-stack web development, frontend-backend integration, database management, API development, authentication systems, testing, debugging, and software project documentation. The overall experience significantly improved my technical and professional skills.

TABLE OF CONTENTS

Contents

1	Preface	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
i.	UCT IoT Platform	4
2.2	About upskill Campus (USC)	8
2.3	The IoT Academy	10
2.4	Objectives of this Internship program.....	10
2.5	Reference.....	10
2.6	Glossary	11
3	Problem Statement.....	12
4	Existing and Proposed solution.....	13
4.1	Code submission (Github link).....	13
4.2	Report submission (Github link).....	14
5	Proposed Design/ Model	15
5.1	High Level Diagram (if applicable)	15
5.2	Low Level Diagram (if applicable).....	16
5.3	Interfaces.....	17
6	Performance Test.....	19
6.1	Test Plan/ Test Cases	19
6.2	Test Procedure.....	19
6.3	Performance Outcome	20
7	My learnings.....	21
8	Future work scope	22

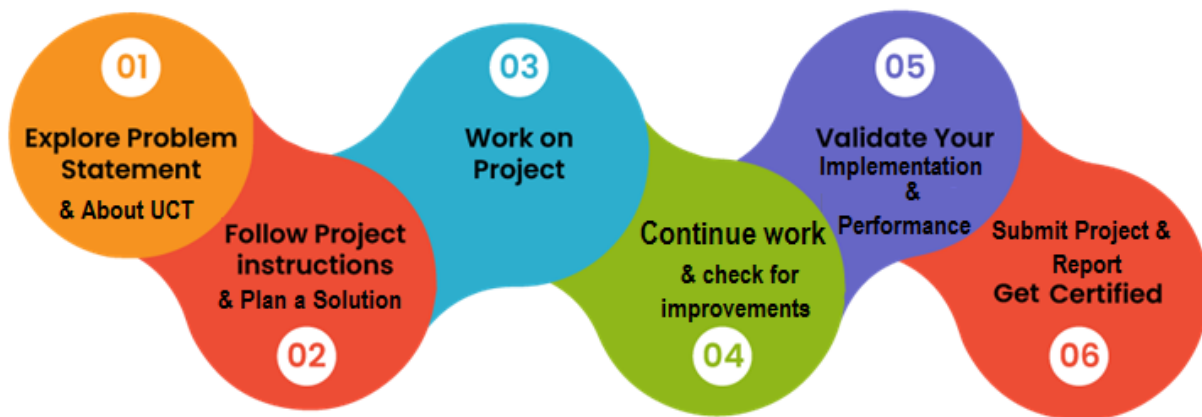
1 Preface

The Industrial Internship organized by upskill Campus and UniConverge Technologies Pvt. Ltd. provided an excellent opportunity to gain practical exposure to software development and industry-oriented project execution.

During this six-week internship, I worked on a Full Stack Web Development project named "ServiceHub". The project focused on developing a service marketplace platform where service providers can list their services and customers can browse and book them through a secure online system.

The internship helped me understand the complete software development lifecycle, including requirement analysis, system design, frontend development, backend development, database management, testing, debugging, and documentation.

I would like to express my sincere gratitude to upskill Campus, UniConverge Technologies Pvt. Ltd., mentors, faculty members, and everyone who supported me throughout this internship. This experience enhanced my technical skills and increased my confidence in developing real-world software solutions.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



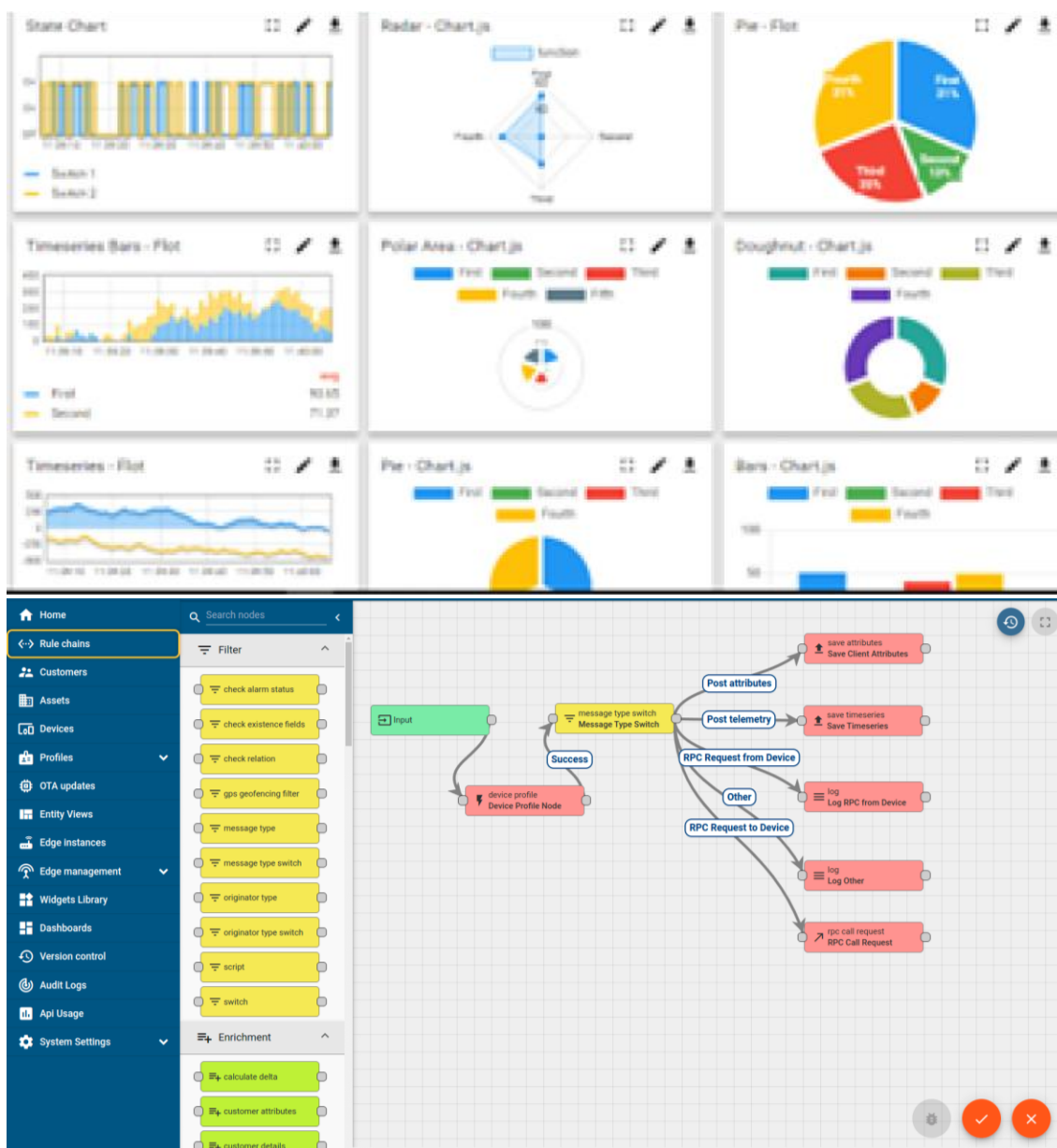
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



FACTORY WATCH

ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



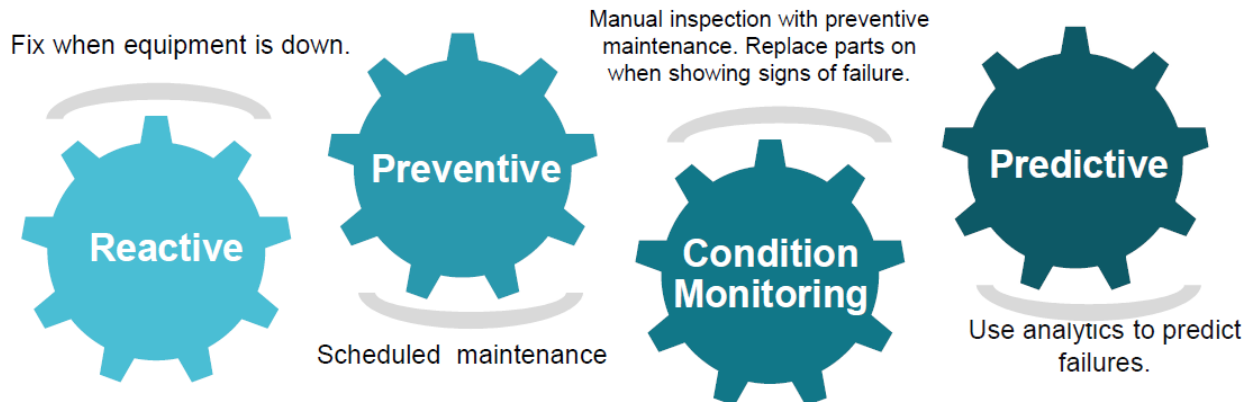


iii. based Solution

UCT is one of the early adopters of LoRAWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

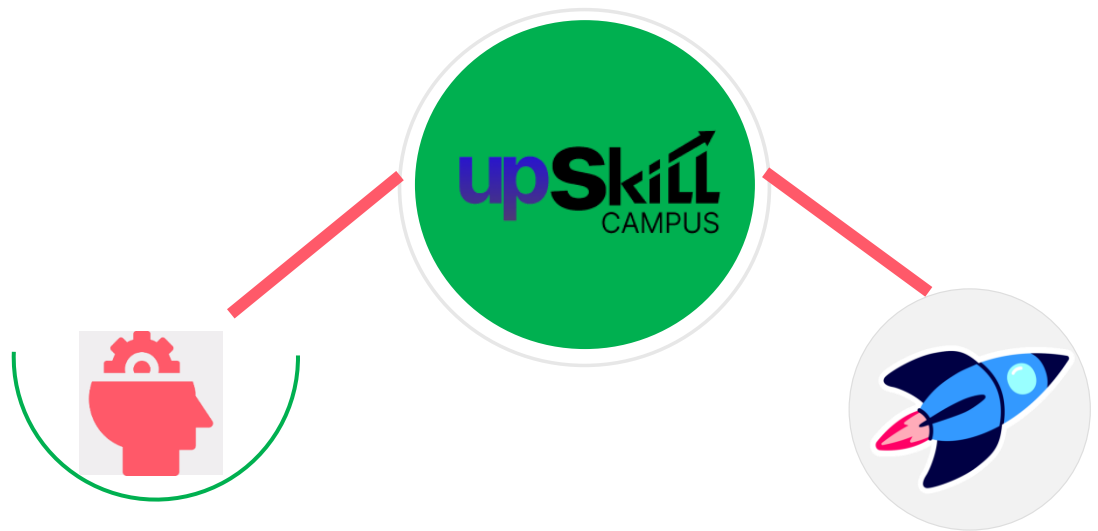
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

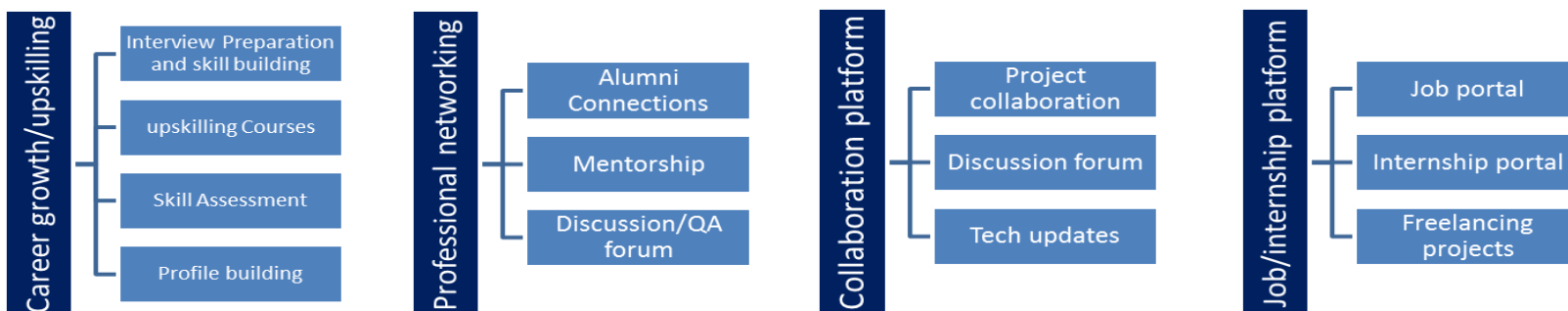
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objectives of this internship project were:

- To gain practical experience in full-stack web development.
- To understand industry-standard software development practices.
- To develop a complete MERN stack application from scratch.
- To implement secure authentication and authorization using JWT.
- To design and integrate REST APIs for frontend-backend communication.
- To develop a service marketplace platform connecting customers and service providers.
- To improve problem-solving, debugging, and software testing skills.
- To gain experience in version control and project management using Git and GitHub.

2.5 Reference

- [1] React.js Documentation – <https://react.dev>
- [2] Node.js Documentation – <https://nodejs.org>
- [3] Express.js Documentation – <https://expressjs.com>
- [4] MongoDB Documentation – <https://www.mongodb.com>
- [5] GitHub Repository – <https://github.com/tusharnigam2005/upskillcampus>
- [6] JWT Documentation – <https://jwt.io>

2.6 Glossary

Terms	Acronym
MERN	MongoDB, Express.js, React.js, Node.js
JWT	JSON Web Token
REST API	Representational State Transfer Application Programming Interface
CRUD	Create, Read, Update, Delete
UI	User Interface
DB	Database
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation

3 Problem Statement

The rapid growth of local service businesses has increased the need for a centralized digital platform where customers can easily discover and book services while service providers can effectively manage and promote their offerings.

Traditionally, customers rely on personal recommendations, social media platforms, or local advertisements to find service providers. This process is often time-consuming, inefficient, and lacks reliability. Similarly, service providers face challenges in reaching potential customers and managing service requests efficiently.

The objective of the ServiceHub project is to address these challenges by developing a web-based service marketplace that enables customers to browse available services, register securely, and book services online. The platform also allows service providers to list and manage their services through an intuitive interface.

The proposed solution aims to improve accessibility, streamline service management, and provide a secure and user-friendly experience for both customers and service providers.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Existing methods for discovering local services primarily depend on offline communication, social media advertisements, and fragmented online platforms. These approaches often lack centralized management, secure booking systems, and efficient communication between customers and service providers.

Some existing platforms are limited to specific categories of services and may not provide a simple mechanism for service providers to manage their offerings. Additionally, users often face difficulties in comparing services, maintaining booking records, and accessing reliable information.

What is your proposed solution?

ServiceHub is a full-stack web application developed using the MERN stack to provide a centralized service marketplace platform.

The system enables customers to register, browse available services, and book services online. Service providers can securely log in, add new services, manage service listings, and monitor customer bookings.

The application incorporates JWT-based authentication, role-based access control, REST APIs, and MongoDB database integration to ensure secure and efficient operation. The responsive user interface developed using React.js provides a seamless experience across multiple devices.

By integrating service discovery, service management, and booking functionality into a single platform, ServiceHub simplifies the interaction between customers and service providers.

What value addition are you planning?

User Authentication	Secure access control
Service Listing	Easy service management
Online Booking	Simplified booking process
MongoDB Integration	Efficient data storage
REST APIs	Smooth frontend-backend communication
Responsive UI	Better user experience

4.1 Code submission (Github link)

GitHub Repository:

<https://github.com/tusharnigam2005/upskillCampus>

4.2 Report submission (Github link)

GitHub Repository:

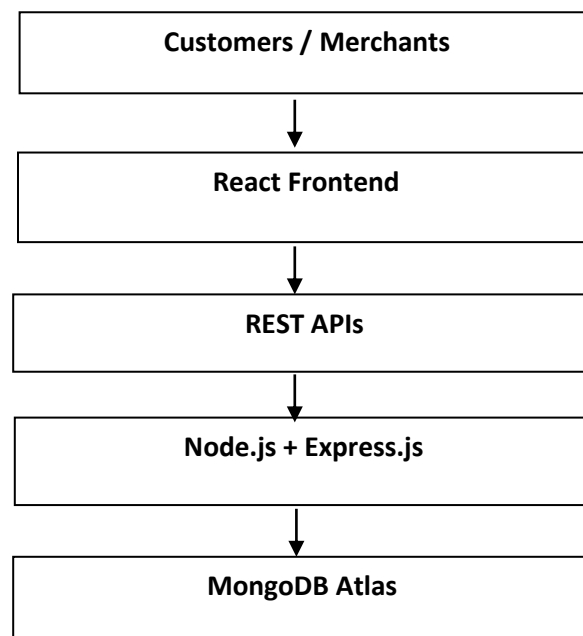
<https://github.com/tusharnigam2005/upskillCampus>

5 Proposed Design/ Model

The ServiceHub application follows a client-server architecture developed using the MERN stack. The frontend is developed using React.js and communicates with the backend through REST APIs. The backend is built using Node.js and Express.js and handles authentication, service management, and booking operations. MongoDB Atlas is used as the cloud database for storing user, service, and booking information.

JWT authentication is implemented to secure user access and role-based authorization is used to provide different functionalities to customers and service providers. The overall architecture ensures scalability, maintainability, and efficient communication between system components.

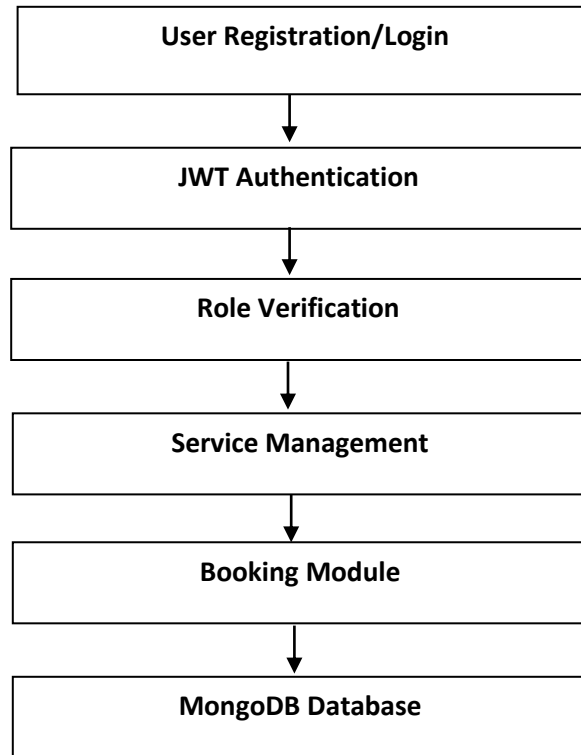
5.1 High Level Diagram (if applicable)



High Level Diagram Architecture

Figure 5.1 shows the high-level architecture of the ServiceHub system. Users interact with the React.js frontend, which communicates with the backend through REST APIs. The backend processes requests and stores application data in MongoDB Atlas.

5.2 Low Level Diagram (if applicable)



The low-level design represents the internal workflow of the application. Users first register and authenticate using JWT-based authentication. Based on their role, the system grants access to service management and booking functionalities. All data is securely stored and managed in MongoDB Atlas.

5.3 Interfaces

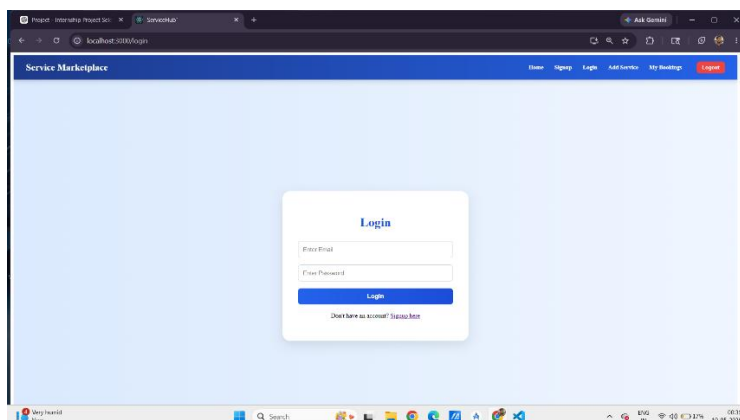


Figure 5.3.1 Login Interface

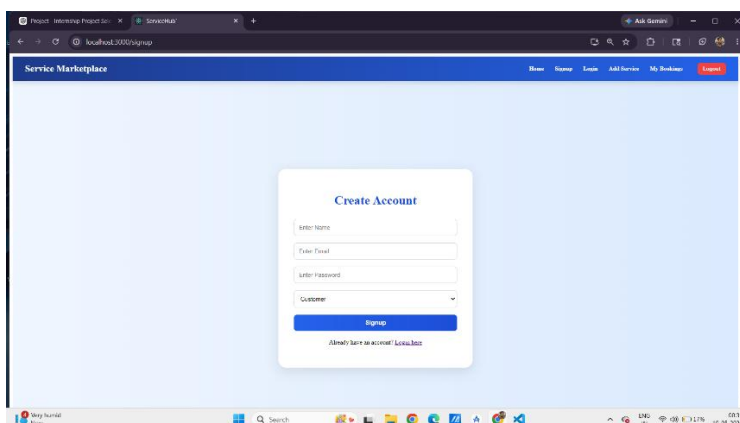


Figure 5.3.2 Signup Interface

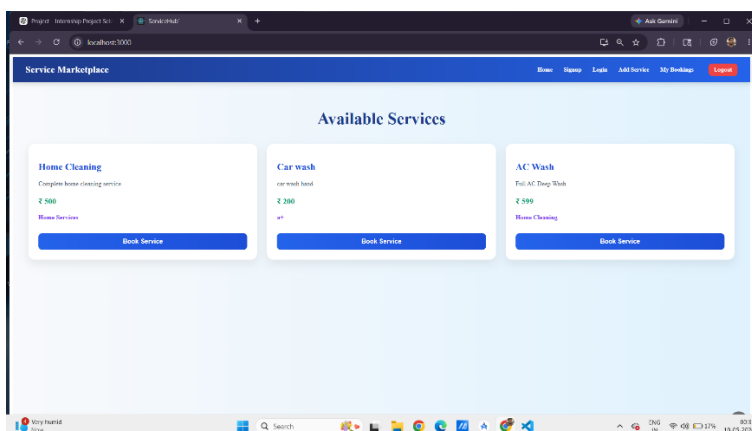


Figure 5.3.3 Services Interface

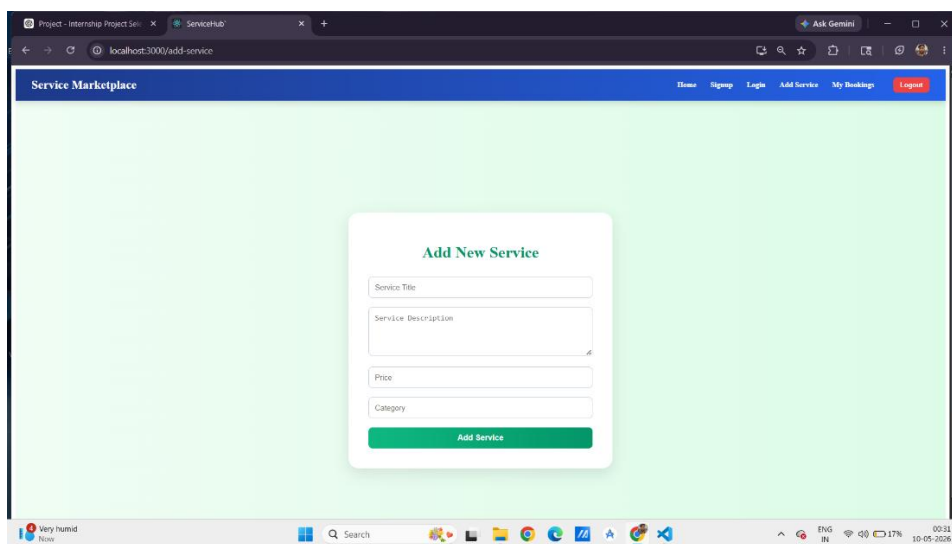


Figure 5.3.4 Add Service Interface

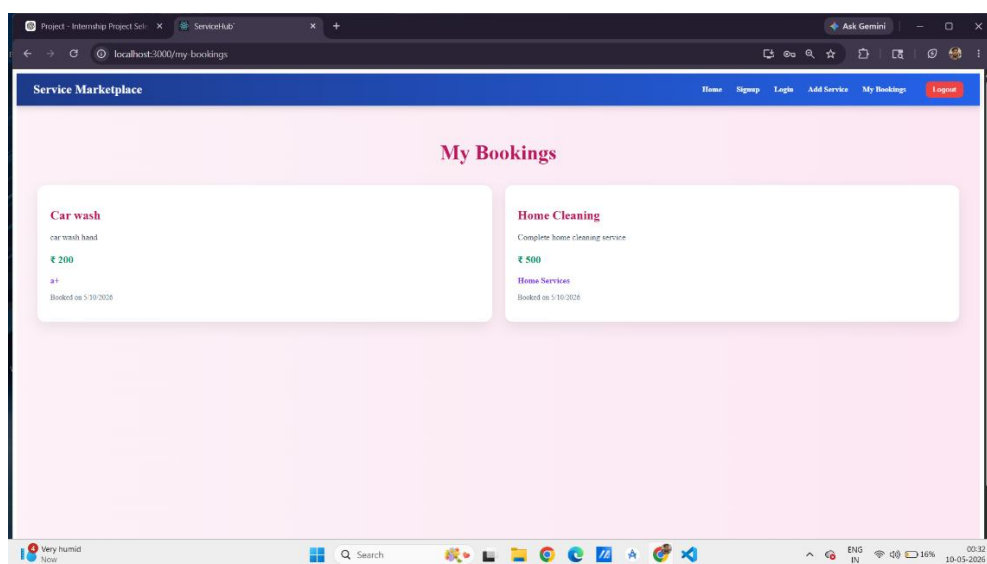


Figure 5.3.5 Booking Interface

The user interfaces of ServiceHub were designed to provide a simple and intuitive experience. The application includes secure login and registration screens, service browsing pages, service management functionality, and booking management interfaces. React.js was used to create responsive pages that work efficiently across different devices and screen sizes.

6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Expected Result	Status
TC-01	User Registration	User account created successfully	Passed
TC-02	User Login	JWT token generated successfully	Passed
TC-03	Add Service	Service added to database	Passed
TC-04	View Services	Services displayed correctly	Passed
TC-05	Book Service	Booking successfully stored	Passed
TC-06	View Bookings	Booking history displayed	Passed

6.2 Test Procedure

The application was tested using both frontend and backend validation techniques. Backend APIs were verified using Postman to ensure correct request handling and database interaction. Frontend functionality was tested through the React application to verify navigation, authentication, service management, and booking operations.

The testing process included user registration, login validation, service creation, service browsing, and booking workflows. MongoDB Atlas was monitored to verify successful data storage and retrieval.

6.3 Performance Outcome

The ServiceHub application performed successfully during testing. User authentication, service management, and booking functionalities operated as expected. REST API communication between the frontend and backend was reliable, and MongoDB Atlas stored and retrieved data efficiently.

The responsive user interface provided a smooth user experience across different screen sizes. Overall, the project met the intended objectives and demonstrated stable performance under normal operating conditions.

The application maintained stable performance during testing. API responses were received within acceptable time limits. MongoDB Atlas efficiently handled user, service, and booking data operations.

7 My learnings

During this internship, I gained practical experience in full-stack web development using the MERN stack. I learned how to build responsive user interfaces using React.js, create backend APIs using Node.js and Express.js, and manage data using MongoDB Atlas.

I also gained hands-on experience with JWT authentication, REST API development, GitHub version control, project documentation, debugging, and software testing. Working on ServiceHub improved my understanding of real-world software development workflows and strengthened my problem-solving abilities.

The internship helped me develop both technical and professional skills, including project planning, requirement analysis, and effective implementation of software solutions.

8 Future work scope

Although the current version of ServiceHub successfully provides service listing and booking functionality, several enhancements can be implemented in future versions.

Future improvements may include payment gateway integration, service ratings and reviews, real-time notifications, advanced search and filtering, admin dashboards, cloud deployment, and mobile application support. Artificial intelligence-based service recommendations can also be incorporated to improve user experience and service discovery.

These enhancements would further improve scalability, usability, and overall platform effectiveness.